In week 4 I choose Quicksort because I was interested in how to most common search algorithm works and when I searched between 2234-08-09T09:00:00 and 2234-08-09T10:00:00 I found 4 entries and the binary search took 0.001 ms and the binary search tree took 0.1 ms. That time isn't completely accurate because that time includes the time it takes to put everything in a tree and then searching the range. In week 5 I implemented a tree data structure. My Tree script includes nodes, where each node stores an employee object and references to two child nodes, a left child and a right child. Each Node can have multiple of the same value and if there are duplicates they are saved in the same node. The search works by checking if the value we have is more or less then the root which is the first node we add. If the value is less than the root we check the left node, otherwise we check the right node and keep going down each side of the tree and create new nodes when needed. The range search works by using a recursive function that gets the lower and upper value. The function checks if the current node is between the lower and upper bounds, if the value is bigger then the lower bound the left node is searched and if the value is smaller than the upper bound the right value is searched. The current node is added to the list if it fits between the low and high values. When the function stops the list is returned and displayed on a scriptable object.

One thing that was a bit of a challenge was visually understanding recursive functions and being able to implement them. One other thing that I have learnt during this course is using namespaces and classes to be able to use functions and classes in other scripts.